

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	Date: 18/07/2026	Enrolment No:92400133050

AIM: Simulate linear convolution and circular convolution on discrete time signals.

Code :

```

import numpy as np
import matplotlib.pyplot as plt

def linear_convolution(signal1, signal2):
    # Compute the linear convolution
    linear_conv = np.convolve(signal1, signal2, mode='full')
    return linear_conv

def circular_convolution(signal1, signal2):
    # Compute the circular convolution
    fft_length = len(signal1) + len(signal2) - 1
    fft_signal1 = np.fft.fft(signal1, fft_length)
    fft_signal2 = np.fft.fft(signal2, fft_length)
    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)
    return circular_conv

# Define the discrete-time signals
signal1 = np.array([2, 2, 4, 4, 6])
signal2 = np.array([1, 3, 6, 9, 12])

# Compute the linear convolution
linear_conv = linear_convolution(signal1, signal2)

# Compute the circular convolution
circular_conv = circular_convolution(signal1, signal2)

# Plot the linear and circular convolution results
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(linear_conv)
plt.title('Linear Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')

plt.subplot(2, 1, 2)
plt.stem(circular_conv)
plt.title('Circular Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')

plt.tight_layout()
plt.show()

# Save the linear and circular convolution results (optional)

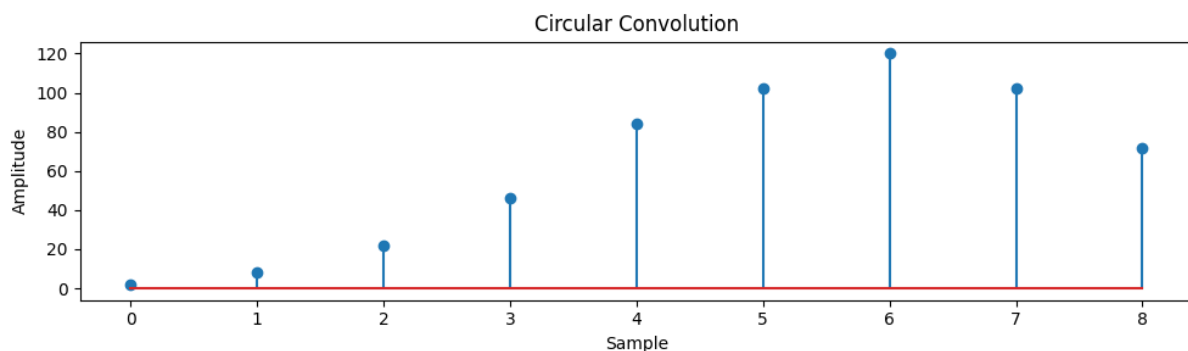
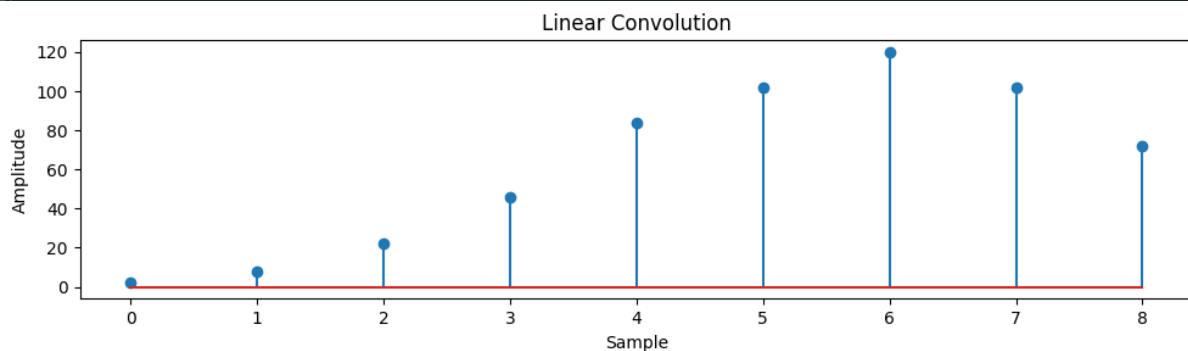
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	Date: 18/07/2026	Enrolment No:92400133050

```
# np.savetxt('linear_convolution.txt', linear_conv, delimiter=',')
# np.savetxt('circular_convolution.txt', circular_conv, delimiter=',')
```

Output :

Name	Type	Size	Value
circular_conv	Array of complex128	[9]	[2.-3.15796771e-15j 8.+0.00000000e+00j 22.-3.15796771e-15j 46.+ ...
linear_conv	Array of int64	[9]	[2 8 22 46 84 102 120 102 72]
signal1	Array of int64	[5]	[2 2 4 4 6]
signal2	Array of int64	[5]	[1 3 6 9 12]



Conclusion :

In this experiment We learned how to perform **linear convolution** and **circular convolution** on discrete-time signals using Python. The output demonstrated that linear convolution produces the complete response of two signals, while circular convolution performs convolution under periodic (cyclic) conditions using the FFT method.



**Subject: Digital Signal and
Image Processing
(01CT1513)**

**AIM: Simulate linear convolution and circular convolution on
discrete time signals.**

Experiment No: 2

Date: 18/07/2026

Enrolment No: 92400133050

Calculation :

DATE :
PAGE :

Experiment - 2

Q Linear Circular Convolution

Signal 1 = [2 , 2 , 4 , 4 , 6]
Signal 2 = [1 , 3 , 6 , 9 , 12]

	2	2	4	4	6
1	2	2	4	4	6
3	6	6	12	12	18
6	12	12	24	24	36
9	18	18	36	36	54
12	24	24	48	48	72

$y(0) = 2$
 $y(1) = 2 + 6 = 8$
 $y(2) = 4 + 6 + 12 = 22$
 $y(3) = 4 + 12 + 12 + 18 = 46$
 $y(4) = 6 + 12 + 24 + 18 + 24 = 84$
 $y(5) = 18 + 24 + 36 + 24 = 102$
 $y(6) = 36 + 36 + 48 = 120$
 $y(7) = 54 + 48 = 102$
 $y(8) = 72$

$y(n) = [2, 8, 22, 46, 84, 102, 120, 102, 72]$



**Subject: Digital Signal and
Image Processing
(01CT1513)**

**AIM: Simulate linear convolution and circular convolution on
discrete time signals.**

Experiment No: 2

Date: 18/07/2026

Enrolment No: 92400133050

Circular Convolution

$$N = 5 + 5 - 1 = 9$$

$$x_1 = [2, 2, 4, 4, 6, 0, 0, 0, 0]$$

$$x_2 = [1, 3, 6, 9, 12, 0, 0, 0, 0]$$

$y(0)$		2	0	0	0	0	6	4	4	2	1
$y(1)$		2	2	0	0	0	0	6	4	4	3
$y(2)$		4	2	2	0	0	0	6	6	4	6
$y(3)$		4	4	2	2	0	0	0	0	6	9
$y(4)$	=	6	4	4	2	2	0	0	0	0	12
$y(5)$		0	6	4	4	2	2	0	0	0	0
$y(6)$		0	0	6	4	4	2	2	0	0	0
$y(7)$		0	0	0	6	4	4	2	2	0	0
$y(8)$		0	0	0	0	6	4	4	2	2	0

$$y(0) = (2 \times 1) = 2$$

$$y(1) = (2 \times 1) + (2 \times 3) = 8$$

$$y(2) = (4 \times 1) + (2 \times 3) + (2 \times 6) = 22$$

$$y(3) = (4 \times 1) + (4 \times 3) + (2 \times 6) + (2 \times 9) = 46$$

$$y(4) = (6 \times 1) + (4 \times 3) + (4 \times 6) + (2 \times 9) + (2 \times 12) = 84$$

$$y(5) = (0 \times 1) + (6 \times 3) + (4 \times 6) + (4 \times 9) + (2 \times 12) = 102$$

$$y(6) = (0 \times 1) + (0 \times 3) + (6 \times 6) + (4 \times 9) + (4 \times 12) = 120$$

$$y(7) = (0 \times 1) + (0 \times 3) + (0 \times 6) + (6 \times 9) + (4 \times 12) = 102$$

$$y(8) = (0 \times 1) + (0 \times 3) + (0 \times 6) + (0 \times 9) + (6 \times 12) = 72$$

$$y(n) = [2, 8, 22, 46, 84, 102, 120, 102, 72]$$



Marwadi
University

Marwadi University
Faculty of Technology
Department of Information and Communication Technology

**Subject: Digital Signal and
Image Processing
(01CT1513)**

**AIM: Simulate linear convolution and circular convolution on
discrete time signals.**

Experiment No: 2

Date: 18/07/2026

Enrolment No:92400133050